How accurate LIGHT_Kr v0_3 is

The pure-krypton model LIGHT_Kr v0_3, version 0.3.0-kr, on 15 October 2026: v0_2 with its electron cross sections trained on krypton's measured swarm data by machine learning, scored against pure-krypton measurements, with its propagated uncertainty, its reference point and its checks

Bottom line. No measurement in pure krypton covers v0_3's own regime, a gas at 0.1 to 2 atm whose electrons a field heats. The measurements that exist test its parts. Electron swarm measurements measure the cross sections themselves, so v0_3 trains its cross sections on those that agree with one another; every other measurement only scores the model.

- **The electron cross sections, trained.** v0_3 multiplies Biagi's level-resolved cross sections by 7 factors: the momentum transfer at four energies from 0.6 to 20 eV, and the excitation of the four 5s levels, the ten 5p levels and the 37 higher levels. They are fitted to the 8 of krypton's 28 swarm datasets that agree with one another within their errors, through a neural emulator of the solver, an MCMC and a neural posterior estimator [52, 54]. Below 0.2 eV the momentum transfer stays Biagi's: trained there too, it degraded the held-out characteristic energy and 195-K drift velocities, which so became validation sets. On the train sets the drift velocity's error falls from 4.8 to 2.8 % rms and the ionization coefficient's (50 to 300 Td) from 14.0 to 9.9 %. The excitation comes down (the 5s levels $\times 0.89$, the higher levels $\times 0.75$), and two tests that neither the training nor that choice saw improve with it: the ionization coefficient below 50 Td, from -44 to -36 % on average, and the beam's partition, whose energy per ion pair moves from +16 to +10 % and whose excitations per ion pair move from +44 to +24 % of Saito et al.'s measurement [38] (Section 4).
- **What a swarm experiment measures.** A Monte Carlo solution of the same cross sections [3] shows what the two-term solver leaves out: a drift tube times the swarm's bulk drift velocity, 23 % above the two-term's flux one at 300 Td, and a Townsend experiment measures the steady-state coefficient, 9 % below the two-term's there. v0_3 scores the model as the experiments measure it. On the trained set the Monte Carlo gives the same ratios to 1.6 %.
- **Below 50** Td the measured ionization coefficient still lies above the model (-36 % on average), and the 5 datasets there part by up to $\times 4.0$ from one another where they overlap. Every published krypton cross-section set falls below them, which Bordage et al. ascribe to Penning ionization of impurities in the experiments [2]. They are held out of the training by rule, and they cannot be scored as the cross sections' error.
- **Excitation to light, under an electron beam.** At low pump v0_3 turns 0.543 of the deposited energy into 147-nm light. Morozov et al. measured 0.42 ± 0.05 [39], so the error is +29 %. v0_3 makes one photon per excitation and per ion, which is what Saito et al.'s measurement implies (0.547 [38]).
- **At LIGHT's reference point** (1 atm, 300 kW cm^{-3} for 1 μs) v0_3 sends 14.05 % of the input to 147 nm on the kinetic EEDF (v0_2: 14.20 %). Propagated from its sources, that share carries ± 11.8 points in quadrature (± 84 %). The trained cross sections' own posterior adds ± 0.04 points. The largest term, 75 % of the variance, is the excimer's formation, Kr* + 2Kr, whose measurements spread by a factor of 3.2.
- **What decides the answer** is the electrons' 5s $\leftrightarrow$ 5p cycle on Jung et al.'s measured cross sections [10]. The electrons pump Kr* to the 5p rung 34 times for each Kr_2^* formed, and krypton's relaxation of the rung turns 77.7 % of the input into heat.
- **The numerical error is far smaller.** The energy ledger closes to 2×10^{-15} of the input. Halving the kinetic closure's renewal steps or doubling its energy cells moves the 147-nm and near-IR shares by at most 3×10^{-5} of themselves.

Table 1 lists every test, Section 1 reads them as errors and propagates the model's own uncertainty, and the sections after it give each test's detail. The swarm datasets in the role *train* set the cross sections' factors (Section 4); every other measurement in the table only scores the model, and no measurement

set any other rate, which come from the sources in Appendix A.

test	measured	v0_2's error	v0_3's error	role
energy ledger closure, reference point			2×10^{-15} of E_{dep}	bookkeeping
energy balance at every node of the flow network			20 nodes, to 10^{-8}	bookkeeping
drift velocity, the 3 sets that agree [41, 43, 34], 79 points		rms 4.8 %	rms 2.8 %	train
ionization coefficient, 50 to 300 Td, the 5 sets that agree [50, 44, 5, 36, 37], 64 points		rms 14.0 %	rms 9.9 %	train
drift velocity, the 8 sets that do not agree (1953 to 1992)			rms 8 to 47 % by set	held out; the two at 195 K validation
characteristic energy D_T/μ, Koizumi et al. [48], 26 points, 0.005 to 0.9 Td		rms 15.8 %	rms 7.1 %	validation
ionization coefficient below 50 Td, 5 sets [51, 36, 37, 44, 5], 36 points		mean -44 %	mean -36 %	held out by rule
ionization coefficient, Monte Carlo on the same cross sections [3], 20 to 50 Td		+3.7, -4.0, -3.5, -2.5 %		solver check
the Monte Carlo's bulk drift velocity and SST coefficient, trained set against Biagi's file, 30 to 300 Td			within 1.6 %	solver check
energy per ion pair under a beam, from Saito et al. [38]	23.9 eV	+16 %	+10 %	held out
excitations per ion pair, Saito et al. [38]	0.55	+44 %	+24 %	held out
energy per excitation or ion, Saito et al. [38]	15.4 eV	-0.1 %	+1.0 %	the beam's check
VUV efficiency, continuous e-beam, Morozov et al. 2008 [39]	0.42 ± 0.05	+31 %	+29 %	held out
VUV efficiency at 3×10^5 W cm^{-3}, Eckstrom et al. 1988 [40]	near 0.46, their limit	-25 %	-26 % at 1 atm	held out

Table 1: LIGHT_Kr v0_3 against the data, every band at its default, on the kinetic EEDF; v0_2 is the same code on Biagi's file as published. An error is the model less the measurement over the measurement, signed for a single value and rms (or a mean) over a set. A validation set is held out of the training but took part in choosing which factors train (Table 8). The swarm rows compare the model as each experiment measures it: the two-term solution times the Monte Carlo's ratios (Section 4). The beam rows run under the e-beam drive of Section 5.

1 Overall accuracy and its uncertainty

Against the data. Three groups of measurements bear on v0_3, and each tests a part of it (Table 1).

- **The electron swarm** (Section 4) trains the cross sections that give every electron rate and scores what the training did not see. On the 8 sets that agree with one another the drift velocity is within 2.8 % rms and the ionization coefficient within 9.9 % from 50 to 300 Td. Held out, the characteristic energy of Koizumi et al. is within 7.1 % rms (Biagi's file: 15.8 %), a validation rather than a test, as it took part in choosing which factors train. The ionization coefficient below 50 Td, which nothing in the training saw, lies -36 % from the measurements on average, where those part by up to $\times 4.0$ from one another.
- **The e-beam measurements** (Section 5) score the chain from excitation to the 147-nm photon. The trained cross sections move the beam's partition toward Saito et al.'s: +10 % on the energy per ion pair and +24 % on the excitations per pair (v0_2: +16 and +44 %). At low pump the error against Morozov et al. is +29 %, 2.5 of their standard errors. At 3×10^5 W cm^{-3} and 1 atm it is -26 % against

Eckstrom et al.'s report, and it loses 37 % below its own ceiling where their beam lost little.

- **Nothing scores the field-driven cycle.** No pure-krypton measurement in hand tests the 5s $\leftrightarrow$ 5p cycle under a field, which decides v0_3's answer. Its cross sections are measured, and their error is propagated below.

Propagated from the sources. Each input is set to the two ends its source allows and the run repeated, the others at their defaults, on both electron closures (Table 2). The inputs are the 8 bands the rows read and the 3 measured constants with a quoted error that no band covers: $Kr_2^+ + e^-$ ($\pm$10 % [30]), $Kr^+ + 2Kr$ (the spread of the three measurements [26, 27, 28]) and the Kr_2^* lifetime (270 $\pm$ 5 ns [25]). Half the difference between an input's two ends is its term. The trained cross sections are one more input: the reference point is run at 12 draws of the training's posterior on the kinetic EEDF and 40 on the two-temperature closure, and the draws' standard deviation is their term. The quadrature sum treats the terms as independent standard deviations, which a band's range is not, so it is indicative. The excursions from the reference, summed by sign, are the ceiling of the spread, not a bound. The 5p rung's lifetime is not propagated: its light is under 1 % of the input.

On the 147-nm share at the reference point, 14.05 % of the input on the kinetic EEDF, the quadrature sum is $\pm$11.8 points ($\pm$84 %). The excursions sum to -16.6 and +31.3 points. The largest terms are Kr^* + 2Kr (10.27 points), the 5s $\rightarrow$ 5p cross section (3.19) and $Kr_2^* + Kr_2^*$ (3.19). The trained cross sections' term is 0.04 points (0.3 % of the share). On the two-temperature closure the sum is $\pm$12.1 points ($\pm$77 %), the cross sections' term 0.16.

The formation term dominates because Kr_2^* forms in competition with the electrons' pumping of Kr^* to the 5p rung, so a faster formation lets more Kr^* escape the cycle. The default, 1.7×10^{-32} $cm^6 s^{-1}$, is the one measurement made at the pressures the model runs at, 0.5 to 1 atm in an e-beam experiment [19]. The other published values, 1.8 to 4.4×10^{-32}, come from low-pressure afterglows [20, 21, 22], and the band's top is Hughes's 5.5×10^{-32} as Quigley and Hughes print it [23]. Four inputs with no measurement at all are 0 by default. Three of them, $Kr_2^* + Kr_2^*$ and the two stepwise ionizations, each at its band's top cut the share by a fifth to a half. The fourth, the Hornbeck–Molnar share, at its top moves it by -0.03 %: direct and associative ionization make a small part of the ions next to Penning's.

input	its two ends	147 nm, kinetic	147 nm, two-temperature	near IR, kinetic	T_g (K), kinetic
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} to 5.5×10^{-32} $cm^6\ s^{-1}$	14.05 to 34.58	15.78 to 36.55	0.64 to 0.45	808 to 688
$e^- + Kr^* \rightarrow Kr^{**}$, scale	0.5 to 1.4	18.59 to 12.21	20.84 to 13.67	0.58 to 0.67	782 to 819
$Kr_2^* + Kr_2^*$, Penning	0 to 5.0×10^{-10} $cm^3\ s^{-1}$	14.05 to 7.67	15.78 to 8.47	0.64 to 0.69	808 to 846
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	0 to 1.5×10^{-7} $cm^3\ s^{-1}$	14.05 to 8.65	15.78 to 10.87	0.64 to 0.64	808 to 841
$Kr^* + Kr^*$, Penning	1.5×10^{-10} to 5.0×10^{-10} $cm^3\ s^{-1}$	18.32 to 14.05	19.82 to 15.78	0.62 to 0.64	783 to 808
$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	0 to 2.0×10^{-7} $cm^3\ s^{-1}$	14.05 to 11.37	15.78 to 14.04	0.64 to 0.65	808 to 824
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0×10^{-11} to 1.7×10^{-10} $cm^3\ s^{-1}$	15.59 to 14.05	17.30 to 15.78	3.41 to 0.64	783 to 808
$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	$\pm$10 %	13.85 to 14.23	15.60 to 15.93	0.65 to 0.64	810 to 807
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.2×10^{-31} to 3.3×10^{-31} $cm^6\ s^{-1}$	13.98 to 14.26	15.70 to 16.00	0.65 to 0.64	809 to 807
the electron cross sections (trained)	12 and 40 draws of the posterior	14.07 $\pm$ 0.04	15.79 $\pm$ 0.16	0.64 $\pm$ 0.00	808 $\pm$ 0
Kr_2^* lifetime	265 to 275 ns	14.04 to 14.06	15.76 to 15.79	0.64 to 0.64	808 to 808
$Kr(nl) + Kr \rightarrow Kr_2^+ + e^-$, the share	0 to 1	14.05 to 14.05	15.78 to 15.77	0.64 to 0.64	808 to 808

Table 2: The propagated uncertainty's inputs, each alone at its two ends with the others at their defaults, and the outputs at those ends (low end to high end), percent of E_{dep}; for the trained cross sections the mean and standard deviation over the posterior's draws. With every input at its default the 147-nm share is 14.05 % (kinetic) and 15.78 % (two-temperature).

Numerical accuracy.

- **The ledger.** The energy ledger closes to 2×10^{-15} of the input on the kinetic EEDF and to 4×10^{-16} on the two-temperature closure. The energy flows also balance at each of 20 nodes to 10^{-8}: the electron gas, each reservoir, each group of rows and each output.
- **The kinetic EEDF at the reference point.** The run takes 381 kinetic solutions, 276 in the pulse, at E/N from 1.34 to 15.8 Td. Over the pulse's second half the field carries 1.0001 to 1.0002 of the input. There the kinetic and the model's energy balances agree to 9×10^{-6} of the input, and the kinetic mean energy equals the model's to 2×10^{-11}.
- **Convergence.** Half the renewal step moves the 147-nm share by 2×10^{-5} of itself. A renewal at

least every 5 ns moves it by 3×10^{-5}, and twice the energy cells by 3×10^{-5}. A ten times tighter solver tolerance moves it by 1×10^{-7}, the EEDF renewed 381 times against the reference's 381.

- **Restarts.** Where the kinetic EEDF finds no solution from its warm start, the solver starts it again cold. The mean match has one solution from any start, so a restart changes no answer, only the way to it. Over the 41 kinetic runs of this note the solver restarted once, in one run.
- **The change from v0_2.** On Biagi's file as published v0_2 gave the reference point's 147-nm share as 14.20 % (kinetic) and 16.00 % (two-temperature); v0_3 gives 14.05 and 15.78 % (-1.1 and -1.4 %). The pulse's mean T_e moves from 2.24 to 2.28 eV on the kinetic EEDF.
- **The swarm scores** are independent of the grid: half and twice the 1000 energy cells move the train sets' rms by 0.01 points at most. The emulator the training ran on reproduces the solver to 0.10 % in the drift velocity and 0.74 % in the ionization coefficient on designs it was not trained on.

2 What v0_3 is

LIGHT_Kr v0_3 is a 0-D model: one uniform box of pure krypton at constant volume, into whose electrons a given power is put. It follows where that energy goes: into the 5s manifold (Kr^*) and the 5p rung (Kr^{**}), the ions Kr^+ and Kr_2^+, the excimer Kr_2^* and its 147-nm second continuum, the near-infrared 5p $\rightarrow$ 5s lines, heat, and the electron gas. Its energy ledger closes by construction. Every species carries a formation energy, and every row's energy goes to the electrons, the gas, light or storage.

Its network has 85 rows (Appendix A):

- electron-impact excitation of each of the 51 levels of Biagi's cross sections at its own threshold, ionization and elastic loss [1, 6, 7], with the factors trained on the swarm data (Section 4): the four 5s levels make Kr^*, every higher level the 5p rung Kr^{**} (the higher levels cascading through it), the excess over the product's energy heating the gas;
- the Hornbeck–Molnar ionization of the 20 levels at or above Kr_2^+'s energy [4], a band at 0;
- Penning ionization of Kr^* by Kr^*;
- ion conversion $Kr^+ \rightarrow Kr_2^+$, and dissociative recombination of Kr_2^+ onto the 5p rung;
- three-body excimer formation and its 147-nm emission;
- the rung's relaxation on krypton and its near-infrared radiation;
- the electrons' 5s $\leftrightarrow$ 5p cycle on Jung et al.'s measured cross sections [9, 10];
- and, as bands at 0, the rows with no measurement.

The electrons' rates come from the two-term Boltzmann equation with electron-electron collisions [8], solved during the run at the model's own mean energy (the default, minutes a run). The alternative is a Maxwellian with a tabulated tail (the two-temperature closure, seconds a run). A rate known only as a published spread, or not at all, is a band: a default and two ends.

What changed from v0_2. The electron cross sections, and one failure path of the kinetic solver: where no EEDF is found from a warm start, it starts again cold (Section 1), a path v0_2's runs never took. v0_2 read Biagi's level-resolved file as published; v0_3 multiplies it by the 7 trained factors (Table 9): the elastic momentum transfer by a factor read between four energies from 0.6 to 20 eV (1 at and below 0.2 eV), and each group of excitation levels by its own factor. The ionization cross section, which beam experiments measure, is not trained. Every row, band, species and table is v0_2's, and with unit factors v0_3 reproduces v0_2 number for number. At the reference point the 147-nm share moves from 14.20 % to 14.05 % on the kinetic EEDF (-1.1 %) and from 16.00 % to 15.78 % on the two-temperature closure (-1.4 %).

3 The reference point

LIGHT's drive in pure krypton: 1 atm at 300 K, 300 kW cm^{-3} for 1 μs into the electrons, a 10^{15} cm^{-3} seed, 2 μs in all. Table 3 gives its outputs on both closures.

	kinetic EEDF (default)	two-temperature
147 nm, Kr_2^*	14.05	15.78
near IR, 5p $\rightarrow$ 5s	0.64	0.63
heat, reactions and relaxation	83.09	81.37
heat, elastic collisions	2.75	2.74
stored in ions (the seed's ions recombine)	-0.74	-0.74
to Kr^* and Kr^{**} by electron impact, gross	21.05	24.15
to Kr^+	6.26	7.83
to Kr_2^* (formed $\times$ 8.43 eV)	14.23	15.98
Kr^* formed that the electrons pump to 5p (%)	95.7	94.9
Kr^* formed that becomes Kr_2^* (%)	2.8	3.2
5s $\rightarrow$ 5p pumpings per Kr_2^* formed	34	30
T_e, pulse mean (eV)	2.28	2.05
T_g at the end (K)	808	798
n_e at the end (cm^{-3})	4.4×10^{12}	4.6×10^{12}

Table 3: The reference point at t = 2 μs, percent of E_{dep} unless a row says otherwise. The fate rows (the first five, with the other stores) sum to 100; the flows below them overlap.

The electrons pump Kr^* to the 5p rung faster than three-body collisions turn it into Kr_2^*. Krypton relaxes the rung back, and the 1.388-eV gap of every cycle becomes heat: 77.7 % of the input, 47 eV for each 147-nm photon. On the kinetic EEDF the electrons sit at 2.28 eV, and the share differs from the two-temperature closure's by -11 %. The cycle's measured cross section therefore decides the answer, with Kannari's fit in its place as a comparison [11, 12]:

5s $\rightarrow$ 5p cross section	0 (no cycle)	$\times$ 0.5	$\times$ 1, Jung et al.	$\times$ 1.4	Kannari's fit
147 nm, kinetic EEDF	50.6	18.6	14.0	12.2	15.3
147 nm, two-temperature	51.8	20.8	15.8	13.7	17.2

Table 4: The 147-nm share (percent of E_{dep}) against the scale on the measured 5s $\rightarrow$ 5p cross section. The band runs from 0.5, the two metastable levels alone, to 1.4, the measurement's upper error.

Jung et al. measured the excitation out of the two metastable levels, $1s_5$ and $1s_3$, into the ten 5p levels, with stated uncertainties of $\pm$35 and $\pm$40 % [10]. v0_3's Kr^* is the whole 5s manifold, so the two J = 1 levels, for which nothing is measured, take the set of the metastable with the same ion core. The lumped cross section is 1.1×10^{-14} cm^2 at 6 eV (5.5×10^{-15} from the metastables alone). Its Maxwellian rate is $\times$1.40 Kannari's fit at 1 eV, $\times$1.23 at 2 eV and $\times$1.20 at the reference point's 2.28 eV.

Pressure and power move the share as much (Table 5, two-temperature closure). It ranges from a fraction of a percent at 0.1 atm to 61 % at 2 atm and 100 kW cm^{-3}. At 0.1 atm a microsecond of this drive heats the gas to 2043 to 8761 K, beyond the measured range of several rates.

kW cm^{-3}	0.1 atm	0.25 atm	0.5 atm	1 atm	2 atm
100	0.2	1.6	8.6	28.7	61.1
300	0.1	0.5	3.4	15.8	45.4
500	0.0	0.3	2.1	10.9	37.1

Table 5: The 147-nm share (percent of E_{dep}) over pressure and power, 1 μs pulses, the two-temperature closure.

4 The electron swarm, and what the training did

The kinetic EEDF's solver is run as a swarm: no electron-electron collisions, the temporal growth term on, and every grid sized so that the EEDF falls 11 to 25 decades across it. It gives the drift velocity

(the Joule power over eE), the ionization coefficient $\alpha/N = k_i/W$ and the characteristic energy D_T/μ, with the transverse diffusion taken over the same momentum transfer and growth term as the field's flux [8]. Swarm measurements are measurements of the cross sections themselves, so they may train them, while the device and beam measurements of Table 1 may not. v0_3 trains a few factors on Biagi's level-resolved cross sections against the swarm data that agree with one another, and scores the rest.

The data. 28 datasets, 441 points, from Dutton's survey [35] and LXCat's LAPLACE database [6, 7], which digitized them from the papers' tables or figures (Table 6). None prints an error, so each carries a class value: 3 % for a drift velocity measured after 1985, 5 % before, 7 % for a characteristic energy and 10 % for an ionization coefficient. A 2 % error in E/N enters through the model's own slope, which matters where the drift velocity climbs out of the Ramsauer minimum (0.01 to 0.03 Td) and for the ionization coefficient at low field. A dataset's χ^2 per point near 1 means the model agrees with it within these errors. Points above 300 Td are checks only: there the mean energy passes 7 eV, where the two-term approximation and the growth model part ways. The ionization coefficient below 50 Td never trains (below).

dataset	quantity	E/N (Td)	points	role	error (%)	χ^2 per point v0_2	χ^2 per point v0_3	rms (%) v0_2	rms (%) v0_3
Pack et al. 1962, 368 K [34, 35]	W	0.00153–0.00622	5	train	5	0.8	0.6	5	4
Hunter et al. 1988 [41]	W	0.002–3.0	42	train	3	0.8	0.7	3	3
Nakamura 1990 [43]	W	0.298–53	32	train	3	2.9	0.4	6	2
Pack et al. 1962, 300 K [34, 35]	W	0.000791–3.1	24	held out	5	4.5	4.3	12	11
Pack et al. 1992, 300 K [42]	W	0.0008–19	42	held out	3	14.2	17.7	15	15
Pack et al. 1962, 195 K [34, 35]	W	0.00121–0.0924	12	validation	5	3.1	2.8	10	9
Pack et al. 1992, 195 K [42]	W	0.0015–0.0894	10	validation	3	6.1	5.3	9	8
Bowe 1960 [45]	W	0.0247–7.4	53	held out	5	15.6	9.6	25	18
English and Hanna 1953 [46]	W	0.0399–2.0	31	held out	5	86.7	84.7	46	47
Dall'Armi et al. 1992 [44]	W	10–80	15	held out	3	10.3	13.4	11	12
Kucukarpaci and Lucas 1981 [47]	W	14–277	6	held out	5	39.6	48.4	25	26
the same, above 300 Td [47]	W	412–820	4	check	5	2.2	0.3	8	3
Koizumi et al. 1986 [48]	D_T/μ	0.005–0.9	26	validation	7	2.5	0.8	16	7
Al-Amin and Lucas 1987 [49]	D_T/μ	1.4–282	12	check	7	6.6	6.4	21	20
the same, above 300 Td [49]	D_T/μ	565–848	2	check	7	2.9	2.0	11	9
Kruithof 1940 [36, 35]	α/N	51–282	15	train	10	3.3	0.8	19	10
Heylen 1971 [37, 35]	α/N	55–250	21	train	10	0.9	0.2	10	5
Dall'Armi et al. 1992 [44]	α/N	55–80	6	train	10	1.8	0.8	15	11
Jacques et al. 1986 [5]	α/N	60–260	11	train	10	3.2	1.2	17	11
Bhattacharya 1979 [50]	α/N	62–281	11	train	10	0.4	1.3	8	15
Kruithof 1940, below 50 Td [36, 35]	α/N	14–45	11	held out	10	69.3	60.1	71	67
Specht et al. 1980, direct [51]	α/N	18–26	7	held out	10	15.0	10.4	53	46
Dall'Armi et al. 1992, below 50 Td [44]	α/N	20–50	7	held out	10	26.8	21.7	50	47
Heylen 1971, below 50 Td [37, 35]	α/N	28–49	10	held out	10	3.1	1.1	22	14
Jacques et al. 1986, 40 Td [5]	α/N	40–40	1	held out	10	0.0	1.2	2	17
Specht et al. 1980, total [51]	α/N	4.5–24	12	check	10	511	507	92	91
the same, above 300 Td [50]	α/N	312–625	4	check	10	1.1	2.7	12	19
the same, above 300 Td [36, 35]	α/N	339–1017	9	check	10	0.3	0.4	5	7

Table 6: Krypton's electron swarm data, their roles and the model's error on Biagi's file as published (v0_2) and on the trained set (v0_3), each as the experiment measures it. Drift velocities W at 300 K unless labelled; Hunter et al.'s at 301 K. A validation set is held out of the training but took part in choosing which factors train (Table 8).

What a swarm experiment measures. The two-term solution with the temporal growth term gives the flux drift velocity and k_i/W. A drift tube times the swarm's centre of mass, the bulk drift velocity, and a steady-state Townsend experiment measures the spatial growth of its current, the SST coefficient. A Monte Carlo solution of the Boltzmann equation, with no two-term approximation, gives all of them [3]. Run on Biagi's file as published (Table 7), it finds the bulk drift velocity 23 % above the flux one at 300 Td, and the SST coefficient 9 % below the two-term's there. The two-term flux drift velocity matches the Monte Carlo's to 0.3 % from 10 to 300 Td, and from 40 Td up the two-term k_i/W lies 1.4 to 3.9 % below the Monte Carlo's flux coefficient. Below 20 Td, without ionization, the bulk and flux drift velocities

are one, and the two-term drift velocity is compared as it is; its own error there (1.8 % at 0.2 Td) is absorbed by the trained factors, which are effective for v0_3's two-term solver. The two-term transverse diffusion is 46 to 79 % too high from 10 to 300 Td, so D_T/μ is scored only below 1 Td, and Al-Amin and Lucas's characteristic energies, 1.4 Td and up [49], are a check of the correction rather than of the cross sections. v0_3 compares the two-term solution times these ratios with each measurement. The ratios are taken on Biagi's file; on the trained set the Monte Carlo gives the drift velocity's and the ionization coefficient's to 1.6 %. Without them the model would be scored, and trained, against the wrong quantity: v0_2 compared its two-term coefficient with Townsend experiments, which put it closer to them at high field than it is.

	Biagi's file			the trained set	
E/N (Td)	W_{bulk}	α_{SST}	D_T/μ	W_{bulk}	α_{SST}
20	0.999	1.036	0.63	–	–
30	1.003	1.036	0.60	1.002	1.034
40	1.006	1.036	0.58	–	–
50	1.011	1.025	0.57	1.018	1.008
75	1.038	0.997	0.56	–	–
100	1.065	0.979	0.56	1.070	0.973
150	1.113	0.956	0.59	–	–
200	1.152	0.939	0.62	1.165	0.930
300	1.232	0.906	0.68	1.241	0.908

Table 7: The Monte Carlo's value over the two-term solution's, LoKI-MC at 300 K and 1 Torr with 10^5 electrons [3]. The coefficient below 40 Td takes the 40-Td ratio, where the Monte Carlo counts too few ionizations to resolve it.

The training. The factors multiply Biagi's cross sections: the elastic momentum transfer by a factor read log-linearly between six energies (0.03 to 20 eV, held beyond them), and the excitation of the four 5s levels, the ten 5p levels and the 37 higher levels by one factor each. The factors at 0.03 and 0.2 eV stay at 1 (below), so 7 train. The ionization cross section is not trained: beam experiments measure it, and in a swarm its scale trades against the excitation's. The prior on each factor's logarithm is broad (standard deviation 0.5, a factor 1.65), so the data decide. Each trial of the factors needs the solver at every measured point, so the fits run on a neural network trained to reproduce it: 2000 sets of factors drawn over $\times 0.55$ to $\times 1.82$ (a Latin hypercube [53]), each solved at 81 values of E/N and gas temperature, and a three-layer network fitted to the change of each quantity from Biagi's file (Adam [55]). On designs it was not trained on it reproduces the solver to 0.10 % in the drift velocity, 0.13 % in D_T/μ and 0.74 % in the ionization coefficient, and every answer below is re-checked on the solver itself.

Which data train. The consistency check fits every candidate together, scores each against that joint fit, and drops the worst while its χ^2 per point exceeds 3. Of 16 candidates it drops 8: English and Hanna 1953 (73.7), Kucukarpaci and Lucas 1981 (44.6), Pack et al. 1992, 300 K (15.5), Dall'Armi et al. 1992 (16.4), Pack et al. 1992, 195 K (5.4), Pack et al. 1962, 300 K (5.0), Pack et al. 1962, 195 K (3.6), Bowe 1960 (3.5) (their χ^2 when dropped). The 8 that remain train. The choice of 3 matters little below it: at 2 the same sets remain. At 4, Pack et al. 1962, 195 K and Bowe 1960 join, and the 1960 drift velocities, -9 % from the 1988 ones where both exist on average, pull the momentum transfer at 6 eV to $\times 1.31$ and the 5s excitation to $\times 1.16$.

Which factors train. Two of the nine stay at 1, by a choice made on held-out data (Table 8). With all nine free the fit matches the train sets best (χ^2 0.49 per point pooled, against 0.64), but it lowers the momentum transfer at the Ramsauer minimum to $\times 0.88$ and raises it at 0.03 eV to $\times 1.13$, a trade the train sets allow and the data that weigh those energies differently reject: Koizumi et al.'s characteristic energy [48] goes from χ^2 2.5 per point on Biagi's file to 8.1, and Pack et al.'s drift velocities in gas at 195 K [34, 42] from 3.1 to 5.5 (1962) and from 6.1 to 8.7 (1992). With the factors at 0.03 and 0.2 eV held at 1, every train set stays within the consistency threshold (Hunter et al.'s drift velocity at 0.7 rather than 0.1), and those three fit better than on Biagi's file, at 0.8, 2.8 and 5.3, the best of the five variants. Holding the 0.6-eV factor too, or every momentum-transfer factor, fits both the train sets and these

three worse. v0_3 trains that variant's 7 factors. Because these held-out sets chose it, they validate the training rather than test it. The tests that neither the training nor this choice saw are the ionization coefficient below 50 Td and the beam's partition (Section 5).

	train sets			held out (validation)		
factors trained	pooled	worst	Hunter 1988	Koizumi 1986	Pack 1962, 195 K	Pack 1992, 195 K
none (Biagi's file as published)	1.75	3.28	0.79	2.52	3.07	6.12
the three excitations	1.16	2.35	0.79	2.52	3.07	6.12
these, and the momentum transfer at 2 to 20 eV	0.66	1.26	0.75	2.43	2.89	5.29
these, and at 0.6 eV too (v0_3)	0.64	1.25	0.70	0.84	2.77	5.27
all nine	0.49	1.27	0.13	8.06	5.45	8.70

Table 8: Which factors train: each variant's least-squares factors on the train set, and the χ^2 per point on the solver itself, each dataset as its experiment measures it with Table 6's errors. Pooled is over the train sets' 143 points. v0_3 trains the fourth row's factors.

The fit. Least squares on the train set gives the factors; an ensemble MCMC [52] (acceptance 0.43) gives their posterior, and a neural posterior estimator, a network trained on simulated datasets to read the factors off them [54], gives the same answer independently (Table 9). Its $\pm 1\sigma$ intervals cover 0.60 to 0.65 of held-out simulations (ideal 0.68). The data pin the momentum transfer from 0.6 to 20 eV and the 5s and higher excitations (the network's R^2 at least 0.83); they barely pin the 5p excitation (R^2 0.15).

factor on	least squares	MCMC median	16–84 %	network	R^2
momentum transfer at 0.03 eV	1, not trained (Table 8)				
the same at 0.2 eV	1, not trained (Table 8)				
at 0.6 eV (the Ramsauer minimum)	1.157	1.157	[1.062, 1.260]	1.125 ± 0.09	0.93
at 2 eV	0.926	0.926	[0.891, 0.962]	0.909 ± 0.04	0.99
at 6 eV	1.039	1.040	[1.008, 1.071]	1.038 ± 0.03	0.99
at 20 eV and above	1.227	1.221	[1.106, 1.354]	1.240 ± 0.10	0.91
the four 5s levels (1s)	0.886	0.871	[0.776, 0.963]	0.838 ± 0.12	0.91
the ten 5p levels (2p)	0.925	0.954	[0.693, 1.293]	1.047 ± 0.34	0.15
the 37 higher levels	0.753	0.738	[0.631, 0.854]	0.703 ± 0.17	0.83

Table 9: The trained factors on Biagi's cross sections. v0_3 runs on the least-squares values. The network column gives its mean and the standard deviation of the factor's logarithm; R^2 is the network's on held-out simulations.

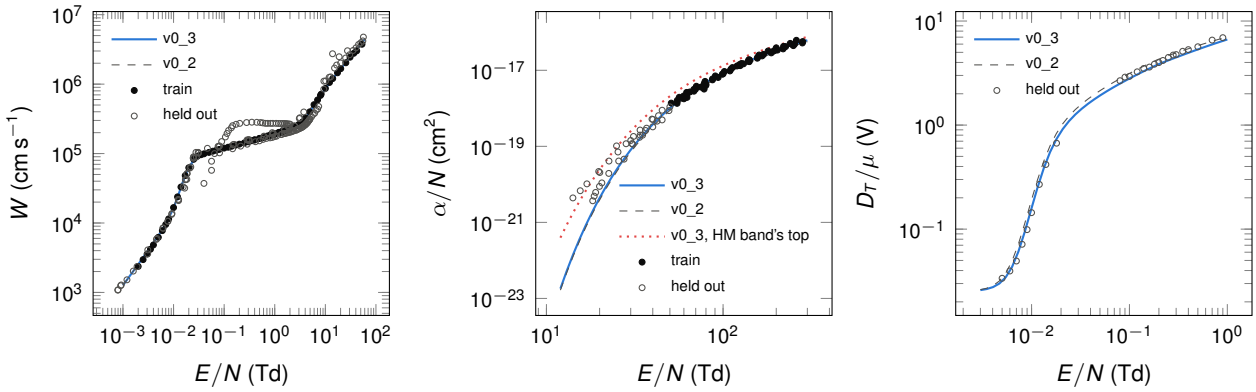


Figure 1: Krypton's drift velocity (left), first Townsend coefficient (middle) and characteristic energy (right) at 300 K, as the experiments measure them: v0_3 on its trained cross sections (solid), v0_2 on Biagi's file as published (dashed), and the measurements of Table 6, filled where they train and open where they are held out (the validation sets among them). Middle: the dotted curve is v0_3 with every excitation of the levels at or above Kr_2^+'s energy ionizing associatively (the Hornbeck–Molnar band's top).

The scores.

- **The train sets.** The drift velocity of Hunter et al. [41], the most precise below 3 Td, moves from χ^2 0.8 to 0.7 per point, Nakamura's [43] from 2.9 to 0.4 (mean +5 to 0 %). The ionization coefficients of Jacques et al. [5] and Kruithof [36], which lay −15 and −18 % from the model, now lie −5 and −7 %; Bhattacharya's [50], which v0_2 matched (−1 %), now lies +11 %. Every train set ends within the consistency threshold.
- **The held-out drift velocities** are the sets the check dropped. The training moves 5 of the 8 closer to the model, the two at 195 K among them, which took part in choosing which factors train. Three move further: Pack et al. 1992 at 300 K [42] (χ^2 14.2 to 17.7 per point), Dall'Armi et al.'s [44] (10.3 to 13.4) and Kucukarpaci and Lucas's [47] (39.6 to 48.4, −17 % from the model on average). Pack et al.'s 1992 set scatters by up to 30 % against the model between points within 5 % of each other in E/N; Bowe's [45] lie +16 % from the model, and English and Hanna's [46] −18 %.
- **The characteristic energy** of Koizumi et al. [48], held out, probes the momentum transfer near the Ramsauer minimum, which the drift velocities pin least: rms 15.8 % on Biagi's file and 7.1 % on the trained set. It chose which factors train, so it validates the training rather than tests it.
- **Below 50** Td, held out by rule, the mean error moves from −44 to −36 %: Heylen's from −20 to −9 %, Specht et al.'s direct coefficient [51] from −52 to −46 %, Kruithof's from −67 to −63 %.

Why the low field stays out. At 30 Td the electrons excite krypton 364 times for every ionization, so anything that ionizes an excited atom, or frees an electron at a surface with its light, can dominate the measured current. Specht et al. found the coefficient anomalously large at low field and attributed it to photoelectron emission at the surfaces by radiating metastables, which they say also explains why earlier measurements disagreed with transport calculations; their direct coefficient leaves that contribution out [51]. Bordage et al. found every published krypton set below the low-field data and ascribe it to Penning ionization of impurities [2]. And the 5 datasets part by up to ×4.0 from one another where they overlap. Training on them would fit the cross sections to the purity and geometry of each experiment. Pure krypton has one process there of its own, the associative (Hornbeck–Molnar) ionization of atoms excited above 12.85 eV [4]. At the band's top it raises the trained set's coefficient ×12.7 at 14 Td, ×3.1 at 30 Td and ×1.5 at 100 Td, and puts the model +54 % from the train sets on average, so a share that large cannot hold where those are measured. No krypton measurement gives the share, and it stays at 0 by default.

What this means for the reference point. The electrons sit at 1.34 to 15.8 Td. There direct ionization makes 0.02 % of the ions and Penning ionization the rest, so the low-field coefficient does not reach the answer. The excitation rates do, and the training moves them: the 147-nm share goes from 14.20 to

14.05 %, and the posterior of the factors spreads it by ± 0.04 points (Section 1).

5 Under an electron beam

Three measurements bear on the chain from excitation to the 147-nm photon. They were made under electron beams, so for them v0_3 runs under a beam drive. The drive is a fixture of this note, not part of the model. It deposits the given power as ion pairs, Kr^*, Kr^{**} and secondary electrons in the proportions of a fast electron's degradation in pure krypton. That degradation is the Fowler–Spencer–Fano recursion on the same cross sections as the model, level by level, from 528 eV: an excitation continues the electron at $E - I$, and an ionization shares $E - I$ equally between the two electrons. Per ion pair it gives:

- W = 26.2 eV;
- 0.410 Kr^{**}, from the levels above the 5s manifold;
- 0.273 Kr^*, from the four 5s levels;
- 0.55 eV of heat, the levels' excess over the energy of the species they make;
- 4.29 eV of sub-excitation secondaries, which the kinetic equation injects with a flat spectrum up to 9.9152 eV.

That is one excitation or ion per 15.55 eV.

The partition. Saito et al. [38] measured scintillation photons and electrons together under 5.49-MeV alpha particles in 1 to 10 atm of krypton. They found 0.55 excited atoms per ion pair, and W_s = 15.4 eV, the energy per photon when every pair recombines. They also confirmed that each recombination gives one photon. On the trained cross sections the drive's 15.55 eV per excitation or ion is +1.0 % from their W_s, its excitations per pair (0.68) differ from theirs by +24 %, and its W differs by +10 % from the 23.9 eV their two numbers imply. On Biagi's file as published (v0_2) the same degradation gave W = 27.6 eV (+16 %) and 0.79 excitations per pair (+44 %): the swarm data's smaller excitation moves both toward the measurement, which neither the training nor the choice of its factors saw. The errors still largely cancel in the quantity the light depends on. One photon of 8.43 eV per 15.4 eV is an efficiency of 0.547. v0_3's ceiling, one photon per excitation and per ion of the drive, is 0.542.

Low pump. Morozov et al. [39] measured the power a continuous 8 to 15-keV beam deposits in the gas against the VUV power emitted into 4π. For krypton they found 0.42 ± 0.05, flat from 500 to 1400 hPa and over a tenfold range of beam power. At 1000 hPa, 30 and 300 W cm^{-3} for 1 μs with the light counted for 50 μs after, v0_3 gives 0.542 and 0.543 (Table 10). The error is +29 %, 2.5 of the measurement's standard errors. At this pump nothing competes with the light: 0.001 Penning ionizations per pair, every ion recombined (1.001 per pair), and every Kr_2^* radiating. So v0_3 sits at its ceiling, which Saito et al.'s measurement supports. Morozov et al. compare their value with calculated maxima of 0.46 and 0.52 for krypton. The 23 % between their value and Saito et al.'s one photon per excitation and per ion (0.547) is not explained in v0_3.

High pump. Eckstrom et al. [40] report the fluorescence efficiency of Kr_2^* near their theoretical limit of about 46 %. They found it under a low-current, long-pulse beam at modest pressures, for pump rates up to 3×10^5 W cm^{-3}. The abstract does not give the pressure of each point. At that pump rate (1 μs, the light counted for 20 μs) v0_3 gives:

- 0.112 at 0.5 atm, 0.342 at 1 atm and 0.523 at 2 atm, that is 79, 37 and 4 % below its ceiling of 0.542;
- at 1 atm, Penning ionization of Kr^* by Kr^* costs 0.625 photons per pair, because the excimer forms too slowly to outrun it;
- at 1 atm with $Kr^* + Kr^*$ at its band's low end, 0.447 (18 % below the ceiling), and with $Kr^* + 2Kr$ at its band's top, 0.504 (7 %).

Their limit assumes other numbers than v0_3's ceiling, so what the report tests is the loss: little at high pump. v0_3 loses little at 2 atm. If their pressures were near 1 atm, the report leans toward a weaker Penning loss or a faster formation than the defaults. Those are the two inputs that also lead the

field-driven uncertainty.

run	measured	kinetic	two-temperature	Penning per pair	photons per pair	T_e (eV)	T_g end (K)
30 W cm^{-3}, 1000 hPa	0.42 ± 0.05	0.542	0.549	0.000	1.682	0.76	300
300 W cm^{-3}, 1000 hPa	0.42 ± 0.05	0.543	0.553	0.001	1.687	0.84	300
3×10^5 W cm^{-3}, 0.5 atm	near 0.46	0.112	0.112	1.337	0.348	0.36	1340
3×10^5 W cm^{-3}, 1 atm	near 0.46	0.342	0.342	0.625	1.063	0.45	688
3×10^5 W cm^{-3}, 2 atm	near 0.46	0.523	0.519	0.072	1.623	0.65	441
1 atm, Kr* + Kr* 1.5×10^{-10}		0.447	0.446	0.301	1.386	0.42	626
1 atm, Kr* + 2Kr 5.5×10^{-32}		0.504	0.502	0.127	1.564	0.53	593

Table 10: v0_3 under the beam drive: the VUV efficiency (147-nm energy over the deposited energy) on both closures, and on the kinetic EEDF the Penning ionizations and 147-nm photons per ion pair, the pulse's mean T_e and the final gas temperature. v0_3's ceiling, one photon per excitation and per ion, is 1.683 photons per pair, an efficiency of 0.542; Saito et al.'s ideal is 0.547.

6 The energy ladder

Figure 2 draws v0_3's levels at their formation energies and the rows that move energy between them.

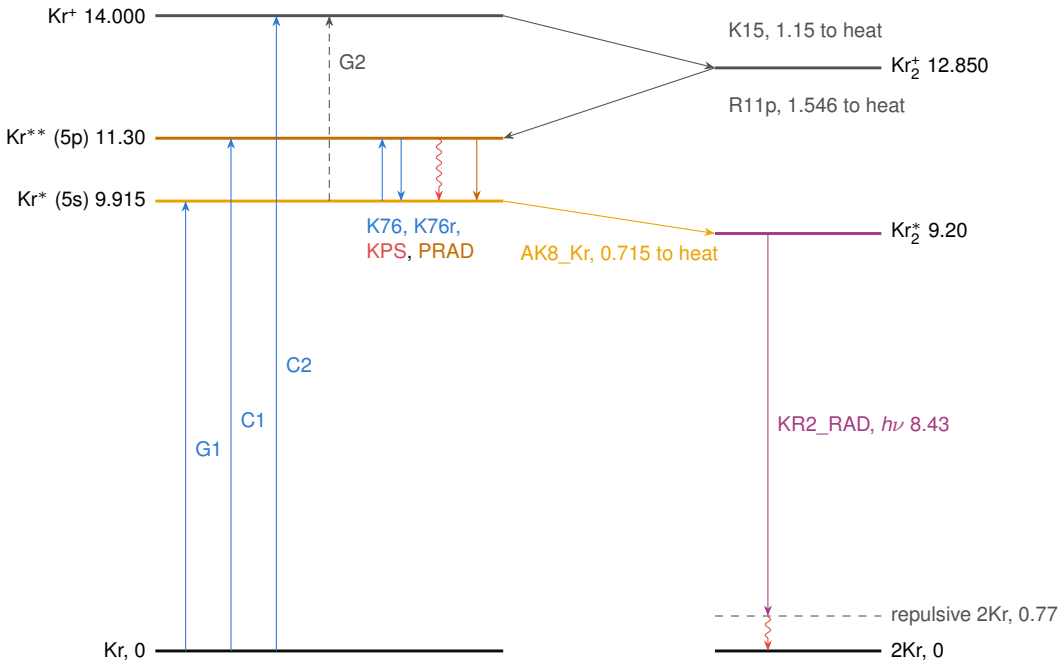


Figure 2: v0_3's energy ladder in eV above ground-state krypton at rest. A bar is a species at its formation energy. Blue arrows are electron impact, which takes its energy from the electron gas; K76r, the 5p → 5s arrow beside K76, returns it. The dashed G2 is Kr* + Kr* → Kr+ + Kr + e$^-$, which gives 5.831 eV to the electron. A red squiggle is heat: KPS relaxes the 5p rung on krypton, and PRAD radiates the 1.388-eV gap in the near infrared. The excimer radiates onto the repulsive ground state, which falls apart and heats the gas. G1's four 5s levels, 9.915 to 10.644 eV, make Kr*; C1's levels, 11.30 to 13.60 eV, cascade to the 5p rung; each heats the gas with its excess. HM, the Hornbeck–Molnar ionization of the levels above Kr_2^+'s energy, is a band at 0 and not drawn.

7 Where v0_3 is not yet reliable

- **Kr* + 2Kr.** The excimer's formation rate is the largest term of the propagated uncertainty: its measurements spread by a factor of 3.2. Its temperature law is measured to 451 K, and the reference

point ends at 808 K.

- **The field-driven regime is unscored.** No pure-krypton measurement in hand tests the 5s $\leftrightarrow$ 5p cycle under a field. Its cross sections are measured to $\pm$35 to 40 %. Those of the two J = 1 levels are not measured; they are the band's low end.
- **The trained cross sections** rest on errors the swarm data do not print (class values here) and on the consistency check's threshold (at 4 rather than 3, two more sets train and the momentum transfer at 6 eV moves to $\times$1.31). The data barely pin the 5p excitation. Which factors train was chosen on held-out data (Table 8), so those sets validate the training rather than test it. The Monte Carlo's ratios are taken on Biagi's file and hold on the trained set to 1.6 %. Their posterior is propagated into the answer ($\pm$0.04 points).
- **The rows with no measurement.** Kr_2^* + Kr_2^* and the stepwise ionizations of Kr^* and Kr^{**} are 0 by default. Each at its band's top cuts the 147-nm share by a fifth to a half (Table 2).
- **Kr^* + Kr^*.** Its 5.0×10^{-10} cm^3s^{-1} is a 1974 measurement [18] as Kannari et al. print it [11]. The e-beam report at high pump leans toward a lower value (Section 5).
- **The e-beam efficiency at low pump** is off Morozov et al.'s by +29 %, and v0_3 holds no process that would explain the difference.
- **The ionization coefficient below 50** Td lies $-$36 % from the measurements on average, which part by up to $\times$4.0 from one another; every published cross-section set falls below them [2]. It is held out of the training by rule, and it does not reach the reference point, where Penning ionization makes the ions.
- **The Hornbeck–Molnar share** of the levels above 12.85 eV is not measured for krypton and is 0 by default. At its top it puts the model +54 % from the train sets and moves the reference point by $-$0.03 %.
- **Processes v0_3 does not carry**, each needing a measured rate before it can enter:
  - Kr_3^+ and its recombination;
  - electron collisions with Kr_2^*: ionization, quenching and the mixing of its two states (v0_3 carries one state at the slow component's 270 ns);
  - Penning ionization between Kr^* and Kr_2^*, and of Kr^{**};
  - the redissociation of Kr_2^+ and Kr_2^* in hot gas;
  - the resonance lines of the two J = 1 5s levels.

  Three-body recombination $Kr^+ + e^- + e^-$ [33] is not carried either: at 1 atm Kr^+ converts to Kr_2^+ in 7 ns.
- **The kinetic EEDF's steady shape.** Between renewals the EEDF keeps its steady shape, which the first nanosecond and the afterglow strain. The two-temperature closure reads a tail table made for LIGHT's Kr/F_2 network on Biagi's four groups (its F_2 = 0 column); the default closure does not use it.

How the numbers were made. Every number in this note is computed from the model when the note is built. None is typed by hand except the measurements, which carry their sources, and the training's record, which the model carries beside its cross sections and the training wrote. The runs are cached under a hash of the model's code, so a change to the model reruns them all (code `f69b10f84305`). The kinetic runs take about 5 minutes each; the two-temperature runs take seconds.

A Reaction rates

The rows of v0_3 with the values they run with and their sources. Electron-impact rows use their cross sections over the kinetic EEDF (or the two-temperature closure). A band gives its default and its two ends.

row	reaction	value	source
G1	$e^- + Kr \rightarrow e^- + Kr^*(5s)$	the four 5s levels, 9.9152 to 10.644 eV, each at its threshold; the excess over 9.9152 to heat	Biagi, Magboltz 8.97 [1, 6, 7], $\times 0.89$ (trained, Table 9)
C1	$e^- + Kr \rightarrow e^- + Kr^{**}$	the 5p, 4d, 6s and higher levels, 11.30 to 13.60 eV, each at its threshold, $\times 0.92$ (5p) and $\times 0.75$ (higher) trained; the excess over 11.30 to heat	Biagi, Magboltz 8.97
HM	$e^- + Kr\ (+ Kr) \rightarrow Kr_2^+ + 2e^-$	the share of the excitations of the 20 levels at or above 12.85 eV; band 0 (default) to 1	no measurement for krypton; the process, Hornbeck and Molnar [4]
C2	$e^- + Kr \rightarrow Kr^+ + 2e^-$	13.9996 eV	Biagi, Magboltz 8.97
G6	e^- + Kr, elastic	the momentum transfer cross section, times the trained factor (Table 9)	Biagi, Magboltz 8.97
K76	$e^- + Kr^* \rightarrow e^- + Kr^{**}$	the lumped measured cross section, 1.1×10^{-14} cm^2 at 6 eV; band $\times$ 0.5 to 1.4	Jung et al. [10, 9]; Kannari's fit as a comparison [11, 12]
K76r	$e^- + Kr^{**} \rightarrow e^- + Kr^*$	detailed balance, g 12/36 over the 1.388-eV gap	
KPS	$Kr^{**} + Kr \rightarrow Kr^* + Kr$	1.7×10^{-10} cm^{3}s^{-1}; band 3.0×10^{-11} to the default	Gazeli et al. [14], one level, products unresolved; the low end lies in the range Chang and Setser measured for argon's 4p levels in argon, 2×10^{-11} to 10^{-10} [16]
PRAD	$Kr^{**} \rightarrow Kr^* + h\nu$, 760–830 nm	τ = 29.0 ns, the $2J + 1$ mean of the ten 5p levels	Dzierżęga et al. [17]; a TALIF lifetime of one level, 34.1 ns, as a cross-check [15]
G2	$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1}; band 1.5×10^{-10} to the default	Leichner and Ericson [18] as Kannari et al. print it [11]; the low end has no published measurement
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} cm^{6}s^{-1} $\times$ $(T_g/300)^{-0.64}$; band to 5.5×10^{-32}	Ueda et al. at 0.5–1 atm [19]; afterglows [20, 21, 22] and Hughes's value [23] inside the band; the law, Wieme and Lenaerts [24]
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$, 147 nm	τ = 270 ns, $h\nu$ = 8.43 eV	Morikawa et al., 270 $\pm$ 5 ns [25]; 250 $\pm$ 20 [20], 280 $\pm$ 30 [23]
PEN_KR2	$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0; band to 5.0×10^{-10}	no measurement; Kannari et al.'s value is the top [11]
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.4×10^{-31} cm^{6}s^{-1} $\times$ $(T_g/300)^{-0.75}$	the mean of three measurements [26, 27, 28]; the law, Pérez-Ríos and Greene [29]
R11p	$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	$1.6 \times 10^{-6}\ (300\,\mathrm{K}/T_e)^{0.55}$ cm^{3}s^{-1}, $\pm$10 %	Shiu and Biondi [30]; Mikus within 8 % [31]
KSTEP_S	$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	0; band to $1.5 \times 10^{-7}\ T_e^{0.68} e^{-4.04/T_e}$ cm^{3}s^{-1}	no measurement; Vriens' calculation [13] as Kannari et al. print it [11]
KSTEP_P	$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	0; band to $2.0 \times 10^{-7}\ T_e^{0.62} e^{-2.53/T_e}$ cm^{3}s^{-1}	the same

Table 11: v0_3's rows. Rates in cm^3 s^{-1} or cm^6 s^{-1}, T_e in eV. Two rows sit in the optional group `electron`, off by default: a stepwise-ionization template of Kr^* and the superelastic return of the 5s levels by detailed balance. Kr_2^+ lies 1.15 eV below Kr^+ + Kr [32]; Kr_2^*'s 0.715-eV well is an estimate that moves energy only between storage and heat.

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